OpenArm-v1 / ExoArm 명령어

Classic CAN 1 Mbps

0 비상 정지 / 초기 disable

```
openarm-can-cli -i can0 disable
openarm-can-cli -i can1 disable
```

1 CAN0 / CAN1 설정

```
sudo ip link set can0 down
sudo ip link set can0 type can bitrate 1000000 restart-ms 100
sudo ip link set can0 up

sudo ip link set can1 down
sudo ip link set can1 type can bitrate 1000000 restart-ms 100
sudo ip link set can1 up
```

2 Classic calibration 스크립트

```
cp /usr/bin/openarm-can-zero-position-calibration ~/zero_calib_classic.py
sed -i 's/oa.OpenArm(args.canport, True)/oa.OpenArm(args.canport, False)/' ~/zero_calib_classic.py
chmod +x ~/zero_calib_classic.py
```

3 오른팔 can0 calibration

```
~/zero_calib_classic.py --canport can0 --arm-side right_arm
```

4 왼팔 can1 calibration

```
~/zero_calib_classic.py --canport can1 --arm-side left_arm
```

5 정상 출력

```
wrote zero positon to arm
[INFO] Motors disabled, exiting safely.
```

6 Sampling check

```
cd ~/openarm_can/build

./openarm-can-motor-sampling-check --all 100 can0
./openarm-can-motor-sampling-check --all 100 can1
```

OpenArm-v1 / ExoArm 명령어

ROS2 Humble

1 터미널 1 - ExoArm ROS2

```
cd ~/foxtech_exoarm/qnbot-exo-std-demo
source /opt/ros/humble/setup.bash
source install/setup.bash
ros2 launch qnbot_teleoperator websocket_teleoperator.launch.py
```

2 터미널 2 - HMI

```
cd ~/foxtech_exoarm/software-tool-v1.2.3/linux
./"Qnbot HMI Control-1.2.3.AppImage"
```

IP HMI forwarding

```
ws://127.0.0.1:19091
```

3 터미널 3 - OpenArm CAN + bridge

```
sudo ip link set can0 down
sudo ip link set can0 type can bitrate 1000000 restart-ms 100
sudo ip link set can0 up

sudo ip link set can1 down
sudo ip link set can1 type can bitrate 1000000 restart-ms 100
sudo ip link set can1 up

source /opt/ros/humble/setup.bash
python3 ~/openarm_teleop/exo_openarm_bridge/exo_to_openarm.py
```

! 비상 정지

```
openarm-can-cli -i can0 disable
openarm-can-cli -i can1 disable
```